

Product Development Roadmap

16 phases · Architecturally ordered · Full implementation detail per phase

KEY DECISIONS IN THIS VERSION

Super Admin CRM moved to Phase 04 — built before the customer CMS

RepoDb + DbUp replaces EF Core — explicit data access, proven migrations

Features spread across phases — each phase ships incremental customer value

16

Phases

3

POS

Systems

5

TV

Platforms

9

Display

Layouts

30+

Feature

Flags

React 18 + TypeScript · .NET 8 Web API · ASP.NET SignalR · RepoDb · DbUp · Azure SQL + Blob

Square · Toast · Clover · Claude AI · Android TV · Fire TV · Samsung Tizen · LG webOS · React Native

Timeline Overview — All 16 Phases

Early phases shown as weeks · Later phases shown as months · Super Admin at Phase 04 — before the customer CMS



01

Foundation & Launch Setup

SETUP · Weeks 1–3 · Milestone: Ready to build

Everything needed before writing product code. Azure at dev scale costs under \$20/mo. These are one-time actions — do them all in week one. The trademark filing is the only item with genuine time pressure: file before any public-facing launch.

Azure Infrastructure — Dev Tier

■ App Service B1 · \$13/mo	Dev/test tier · scales to B2 when first paying customer signs up
■ Azure SQL Basic · \$5/mo	2GB · sufficient for hundreds of venues at launch · upgrade to S1 at scale
■ Azure Static Web Apps · Free	CDN-backed hosting for admin CMS, super admin, and display player SPAs
■ Azure Blob Storage · ~\$1/mo	Menu photos, tap photos, custom display assets, AI-generated HTML templates
■ Azure SignalR Service · Free tier	Up to 20 concurrent screen connections · upgrade to Standard at scale
■ App Insights · ~\$0/mo	Telemetry from day one · free under 5GB/mo · essential for debugging SignalR
■ GitHub repo + Actions CI/CD	Auto-deploy to Azure on push to main · separate staging and production slots

Business Setup

■ Stripe account	2.9% + 30c per transaction · create product and price objects for all 4 tiers immediately
■ Business bank account	Mercury or Relay · free · keeps business finances separated from personal from day one
■ Trademark 'Vennu' — USPTO Class 042	File immediately · \$250–350 + attorney · do not launch publicly before filing
■ Domain registration	vennu.app or chosen alternative · register the moment the name is confirmed — cannot be delayed

Dev Tooling

■ Visual Studio 2022 Community	Free · full .NET 8 IDE · build, debug, breakpoints, NuGet · primary coding environment
■ Vite + React 18 + TypeScript	Lightning-fast HMR · same Vite config across admin CMS, super admin, and display player
■ Claude Code CLI	npm install -g @anthropic-ai/claude-code · Node 18+ required · runs in VS integrated terminal
■ Postman	Free · API testing and webhook simulation · create a Vennu collection from Phase 02 onwards
■ Figma · \$15/mo	UI design · JSX prototypes from planning phase used as production reference

CLAUDE.md. Create CLAUDE.md in the project root before the first Claude Code session. Include: project description, current phase, build commands (dotnet build, dotnet run), naming conventions (screen IDs: sc-{6 random chars}, feature keys: snake_case e.g. happy_hour), and SignalR group format (screen:{screenId}, venue:{venueId}). Claude Code reads this file automatically at the start of every session.

02

Core Backend & Real-Time Engine

BUILD · Weeks 3–7 · **Milestone:** Screens update in real time

The foundation every other phase builds on. Build the .NET 8 API, RepoDb repositories, DbUp migrations, and SignalR hub before any UI. A content save should appear on a test TV within 200ms before Phase 03 begins — validate this end-to-end with two browser tabs.

Data Models

Core entities only — feature flag and tier entities come in Phase 03. All PKs use UNIQUEIDENTIFIER with NEWID() default.

■ Venue	Id, Name, Timezone (IANA string), Type (Restaurant/Bar/QSR/Café/Brewery/FoodHall), PrimaryLanguage, SecondaryLanguage
■ Screen	Id, VenueId, ScreenKey (sc-a3f9bc), Name, Location, WallGroup, WallPosition, LastSeen, Status, Platform, AppVersion
■ ScreenPairingCode	Code (6-digit string), VenueId, ScreenId (null until claimed), ExpiresAt, IsClaimed · TV polls IsClaimed every 3s
■ Menu / MenuSection	Menu: Id, VenueId, Name, IsActive · MenuSection: Id, MenuId, Name, DisplayOrder, IsActive
■ MenuItem	Id, SectionId, Name, Description, Price, HHPrice, Available, Qty, Tags (CSV), ImageUrl, IsPopular
■ MenuItemTranslation	(ItemId, LanguageCode, Name, Description, IsAutoTranslated) · baked in from day one · no schema change in Phase 12
■ TapItem	Id, VenueId, Name, Style, ABV, IBU, Description, Price, GlassColor, NameColor, BreweryName, Available, DisplayOrder

DbUp Migration Setup

■ Install dbup-sqlserver NuGet	Add to Vennu.Data project · no other dependencies
■ Scripts folder	Vennu.Data/Scripts/ · all .sql files set as Embedded Resource in project properties

■ Naming convention	001_create_venues.sql · 002_create_screens.sql · never modify existing scripts, always add new numbered ones
■ WithTransactionPerScript()	Each script runs in its own transaction · failed script rolls back cleanly without affecting already-applied scripts
■ Program.cs startup call	DatabaseMigrator.Run(connectionString) before builder.Build() · API does not accept traffic if any migration fails

RepoDb Setup

■ Install RepoDb.SqlServer NuGet	Add to Vennu.Data · also install RepoDb.SqlServer.BulkOperations for bulk insert and merge operations
■ SqlServerBootstrap.Initialize()	Called in Program.cs · registers all SQL Server type maps and handlers globally
■ Repository pattern	IXxxRepository interface + XxxRepository class per entity · registered as scoped via DI
■ Connection per operation	using var conn = new SqlConnection(_connectionString) · connection opened and closed per call
■ Extension method style	await conn.QueryAsync(s => s.ScreenKey == key) · explicit, readable, no change tracking

REST API — Core Endpoints

Method · Route	Purpose
GET /api/display/{screenId}/content	Full board payload for a screen · called by display player on boot
POST /api/display/{screenId}/heartbeat	Screen health ping every 30s · updates LastSeen and Status
POST /api/screens	Register new screen · returns screenKey (sc-a3f9bc)
POST /api/screens/pairing-code	TV calls this · generates 6-digit code with 10-min expiry
GET /api/screens/pairing/{code}/status	TV polls every 3s · returns {linked, screenId} when admin claims
POST /api/screens/pairing/{code}/claim	Admin claims code · links screen to venue · TV auto-redirects
POST /api/venues	Create venue · assigns default Starter tier

GET /api/venues/{id}/features	Returns computed feature set (tier + overrides) for admin frontend
POST /api/media/upload	Upload image → Azure Blob CDN → returns CDN URL

ASP.NET SignalR Hub — VennuHub.cs

Method / Event	Direction	Purpose
JoinScreen(screenId)	Client → Hub	TV joins its named group on connect · all push events target this group
JoinVideoWall(wallId, position)	Client → Hub	Wall screens join shared group · position used for CSS translateX offset
ContentUpdated(payload)	Hub → Client	Full board payload pushed within ~200ms of any admin save
ThemeUpdated(theme)	Hub → Client	Colours and fonts only · lighter than full ContentUpdated
ItemAvailabilityChanged(itemId, availability)	Hub → Client	Pushed when POS webhook fires · player patches single item in <500ms
SyncTick(serverTimeMs)	Hub → Client	16ms broadcast for video wall frame sync · ~60fps software sync
withAutomaticReconnect()	Client config	Reconnects silently after network drops · no manual page refresh needed

Display Player — Boot Sequence

■ Step 1 — Fetch content	GET /api/display/{screenId}/content · sets initial board state
■ Step 2 — Connect SignalR	JoinScreen(screenId) · subscribes to ContentUpdated, ThemeUpdated, ItemAvailabilityChanged, SyncTick
■ Step 3 — Start heartbeat	POST /heartbeat every 30s · marks screen Online in admin dashboard
■ Step 4 — Register Service Worker	Caches last successful payload · board keeps showing if internet drops mid-service
■ Layout switch	board.layout field selects React component · new layouts = new components only · boot sequence never changes
■ Video wall mode	If wallPosition !== null · full canvas rendered · CSS translateX(-pos × 1920px) clips the correct section

Background Services

■ ScheduleEvaluator	IHostedService · runs every 60s · evaluates all active schedules · pushes ContentUpdated when a change is due
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■ HeartbeatMonitor	Marks screens Offline if no ping in 90s · immediately pushes dashboard notification to admin
■ StripeWebhookReceiver	Handles subscription.updated, invoice.paid, customer.subscription.deleted · all processed asynchronously

03

Tier System & Feature Flags

BUILD · Weeks 6–9 · **Milestone:** Monetisation infrastructure live

Built before any customer-facing UI so `HasFeatureAsync()` already exists when the first admin panel is written in Phase 05. Every feature check, upgrade prompt, and billing webhook in every later phase calls this service.

Data Models

■ Feature	Id, Key (snake_case e.g. 'happy_hour'), Label, Category, IsActive (master kill switch)
■ SubscriptionTier	Id, Name, Slug, Price, MaxScreens (-1 = unlimited), IsPublic, IsActive, StripeProductId
■ TierFeature	TierId, FeatureId, LimitValue (stores metered limits e.g. '20' for AI descriptions per month)
■ VenueSubscription	VenueId, TierId, StripeSubscriptionId, Status (active/trialing/past_due/canceled), TrialEndsAt, CurrentPeriodEnd
■ VenueFeatureOverride	VenueId, FeatureId, Enabled (true=unlock, false=block), Reason (required), ExpiresAt (null=permanent), CreatedByAdminId

Feature Resolution Service

■ HasFeatureAsync(venueId, key)	Checks VenueFeatureOverride first (if not expired) · falls back to tier's TierFeature entries · returns bool
■ GetFeatureSetAsync(venueId)	Returns full computed feature dictionary · used by admin frontend to decide what to show, dim, or lock
■ Override beats tier always	Most specific wins · enables beta testing, custom deals, and grandfathering specific customers
■ Metered features	LimitValue on TierFeature stores numeric cap · <code>GetUsageAsync(venueId, key)</code> tracks monthly consumption
■ IMemoryCache	60s sliding expiry · invalidated on tier change or override update · prevents a DB hit on every request

Initial Tier Definitions

Tier	Price	Screens	Key Features
Starter	\$39/mo	2	photo_grid · classic_diner · basic_scheduling · allergen_badges · analyti
Restaurant Starter	\$49/mo	1	Above + meal_periods · bilingual_display · ai_translation (1 lang) · quick
Pro	\$89/mo	6	All Starter features + all layouts · happy_hour · pos_integration · staff_ap
Business	\$179/mo	∞	All Pro features + ai_custom_builder · multi_location · white_label · html_

14-day free trial with no credit card required. Annual billing: 2 months free (~17% discount). Target 40–60% of customers on annual to improve cash flow.

Stripe Integration

■ Products + Prices per tier	One Stripe product per tier · monthly and annual Price objects · IDs in SubscriptionTier.StripeProductId
■ subscription.created	Sets VenueSubscription.Status = trialing or active · grants full feature access per tier
■ invoice.paid	Confirms payment · extends CurrentPeriodEnd · no action if already active
■ customer.subscription.updated	Handles plan changes · updates TierId · Stripe handles proration automatically
■ customer.subscription.deleted	Sets Status = canceled · feature access restricted immediately
■ Webhook idempotency	All handlers log Stripe event IDs in ProcessedStripeEvents table · safe to receive same event twice
■ HaaS contract billing	Separate Stripe subscription per hardware bundle · early cancel triggers buyout charge calculation

04

Super Admin CRM

BUILD · Weeks 8–11 · **Milestone:** Internal ops tooling ready

Built immediately after the tier system so it can be used during development of the customer CMS. Use it daily during Phases 05–10 to create test venues, assign tiers, toggle feature flags, and validate `HasFeatureAsync()` before any real customer encounters tier logic.

Revenue Dashboard

■ MRR + ARR	Live from Stripe API · month-over-month trend · breakdown by tier
■ Active venue counts	Active · trialing (listed separately) · churned in last 30 days
■ Avg revenue per venue	$MRR \div \text{active venues}$ · tracked over time as a retention health indicator
■ Recent events feed	New signups · upgrades · downgrades · overrides applied · churn · reverse chronological
■ Screen health map	Every screen across every venue as a dot · green=online · red=offline · hover shows venue, screen name, last seen

Tier Manager

■ Create / edit / clone / archive	Full CRUD on tiers · changes take effect immediately · no code deployment needed
■ Feature toggle per tier	Checkbox per feature per tier · Enable All / Clear All shortcuts · save broadcasts to all venues on that tier
■ MaxScreens field	-1 = unlimited · stored as integer · resolver treats -1 as no limit
■ Stripe product ID field	Links tier to Stripe product · validated on save · kept in sync on every tier change
■ Public / private flag	Private tiers invisible on pricing page · used for custom enterprise deals

Feature Matrix

■ All features × all tiers as a grid	One cell = one checkbox · click to toggle · unsaved changes highlighted in amber
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■ **Category grouping**

Display · Scheduling · Language · Mobile · POS · AI · Analytics · Enterprise

■ **Save all changes at once**

Single save button applies all pending toggles in one transaction

■ **Audit trail**

Every change logged with admin ID, timestamp, previous value, and new value

Venue CRM

■ **Search and filter**

Filter by name · tier · status (active/trialing/canceled) · screen health

■ **Venue table**

Name · type · tier · MRR · screen count · last active · override count · health indicator

■ **Tier switcher**

Change any venue's tier in one click · Stripe subscription updated automatically

■ **Override panel**

Add feature unlock or block · reason field (required) · optional expiry date · effective immediately

■ **Effective features view**

Shows each feature, its source (tier default or override), reason if applicable, expiry date

■ **Support context**

See exactly what a customer has access to before picking up any support request

Development workflow. During Phases 05–10: create a test venue per tier in Super Admin, toggle features on and off, and verify the admin CMS shows the correct locked/unlocked state for each. Find tier logic bugs in development rather than after a paying customer signs up.

05

Admin CMS — Core Editing

BUILD · Weeks 10–14 · **Milestone:** First venue can manage their board

The interface restaurant and bar owners use every day. Every panel calls `HasFeatureAsync()` before rendering. Locked features show a tier badge and soft prompt — never an error message. Built mobile-responsive from day one for Quick Update mode.

Menu Editor

■ Section expand / collapse	Manage large menus without scrolling · collapsed state persisted in <code>localStorage</code>
■ Inline item editing	Edit name, description, price, HH price without leaving the list · Save & Sync pushes via SignalR in ~200ms
■ Availability toggle — 86 items	One click marks item unavailable · Live/Off pill · SignalR push in 200ms · auto-resets at midnight
■ Sold-out / limited qty badges	'Only 3 left!' shown on display · auto-removed when Qty reaches zero
■ Allergen and dietary badges	GF · V · Vegan ■ · Halal ■ · Kosher ☆ · Nuts ■ · Spice ■ · stored as CSV on <code>MenuItem.Tags</code>
■ Bestseller tag	★ Popular badge on display board · driven by admin toggle on <code>MenuItem.IsPopular</code>
■ HH price field visibility	Always visible · disabled with PRO badge if below Pro tier · not hidden, shows greyed value

Quick Update Mode

■ Purpose	Built for the solo restaurant owner running lunch service from a phone at the pass
■ Daily special push	Single text input · push button · live on all screens in under 10 seconds
■ 86 toggle list	Every menu item as a large tap-friendly toggle · one scroll · no section navigation
■ Auto-midnight reset	<code>ScheduleEvaluator</code> restores all 86'd items at midnight · no morning action required from any staff

■ Mobile-first layout	Optimised for phones · large touch targets · no horizontal scrolling
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Screen Management

■ Registration URL generation	Admin creates screen record · gets <code>vennu.app/display/sc-a3f9bc</code> · pairing code added in Phase 09
■ Screen health dashboard	Online/Offline per screen · last-seen timestamp · click any screen to push content manually
■ Screen naming and location	'Main Bar TV', 'Patio Board', 'Entrance Display' · location label shown in super admin screen map
■ Push to specific screen	Target one screen or broadcast to all venue screens simultaneously
■ Multi-screen overflow visualiser	Mini strip showing item distribution across screens · density selector (2×2 · 3×2 · 4×2 · 3×3)
■ Video wall builder	Assign screens to wall group by position · 2×1, 3×1, 2×2 configs · Pro tier only

Tier-Aware UI Patterns

Pattern	Behaviour
Tier badge	Small coloured pill (e.g. PRO in amber) beside locked nav item, field, or layout card — informational
Locked nav item	50% opacity in sidebar · tier badge beside label · click opens upgrade modal, not an error
Locked section preview	Blurred glimpse of the feature · one concrete benefit sentence · soft 'Unlock with Pro' CTA · not blocked
Disabled form field	Field visible but greyed · tooltip explains requirement · not hidden
One prompt per screen	Never more than one upgrade suggestion simultaneously · prevents alert fatigue
All prompts dismissible	Per-session dismiss memory · dismissed hint never reappears in the same browser session

06

Display Layouts — Restaurants & Cafes

BUILD · Weeks 13–17 · **Milestone:** Can sell to restaurants and cafes

The first two display layouts cover the broadest market. Photo Grid targets food-photo-heavy venues. Classic Diner is the simple text board most small restaurants actually want. Basic theme builder ships here.

Photo Grid Layout

■ Purpose	QSR · Chinese · Vietnamese · Mexican · any restaurant using food photography as primary menu communication
■ Grid density options	2×2 (4 items) · 3×2 (6 items) · 4×2 (8 items) · 3×3 (9 items) · admin selects per venue
■ Food photo cards	Azure Blob CDN image with gradient placeholder while loading · name, short description, price overlay
■ Bestseller ribbon	★ POPULAR badge top-left of card · driven by MenuItem.IsPopular toggle in admin
■ Sold-out overlay	Card dims to 40% opacity with SOLD OUT text · auto-restores at midnight · instant via SignalR from POS in Phase 11
■ Happy hour pricing	HH price shown in amber · regular price struck through · only when isHappyHour=true in content payload
■ Multi-screen overflow	$\text{start} = (\text{screenPosition}-1) \times \text{itemsPerScreen}$ · screen self-selects its slice · no server-side per-screen logic

Classic Diner Layout

■ Purpose	Diners · cafes · sandwich shops · venues where text legibility at distance matters more than photos
■ White background	Warm cream #faf8f4 · inverts the dark theme · high contrast dark ink typography
■ Multi-column grid	2 or 3 columns · sections side by side · familiar laminated menu aesthetic
■ Category headers	Bold Playfair Display section titles · 1px rule below each · generous vertical spacing

■ Daily special banner	Full-width strip at bottom · 'Soup of the Day: Tomato Bisque · \$5' · driven by Quick Update special field
■ Price alignment	Right-aligned prices with dot leaders · clean and readable at any screen size

Theme Builder — Basic (all tiers)

■ Background colour picker	6 quick swatches + full colour picker · warm cream recommended for Classic Diner
■ Accent colour	Single brand colour applied to prices, section headers, and highlights
■ Venue name font	3 options: DM Sans · Playfair Display · Syne · full library in Phase 07
■ Live preview pane	Shows exact TV output on every change · no save needed to preview
■ Push to all screens	One button · SignalR broadcasts ThemeUpdated to every connected screen instantly

Player Evolution — Phase 06

■ Layout switch added	board.layout selects React component · map grows with each phase · boot sequence unchanged
■ Azure Blob CDN image load	Lazy loaded with gradient placeholder · CDN cache headers set on upload
■ Overflow position logic	screenPos from screen registration · start = (pos-1) × itemsPerScreen
■ Service Worker caching	Full content payload cached · CDN photos cached separately · offline resilient

07

Display Layouts — Bars

BUILD · Weeks 16–20 · Milestone: Can sell to bars and upscale restaurants

The Neon Chalkboard is the visual centrepiece of the platform — the display that makes customers stop and look. It is the primary reason a bar chooses Vennu over a generic signage tool. The full theme builder with neon controls ships alongside it.

Neon Chalkboard Layout

■ Chalk board texture	SVG fractal noise at 5% opacity on near-black · authentic chalk surface
■ Multi-layer neon text-shadow	8-layer CSS stack: white 0px 0px 2px core → coloured mid-glow → wide ambient · true tube-light appearance
■ Neon flicker animation	CSS keyframes on venue title · opacity dips at irregular ms intervals — not a repeating loop
■ Glow breathe animation	Slow 3s ease-in-out brightness pulse on all neon elements · feels alive and warm
■ Chalk draw-in animation	Items clip-path left-to-right on load · as if someone is writing on the board in real time
■ Scanline overlay	repeating-linear-gradient every 4px at 4% opacity · subtle CRT/TV screen texture across the board
■ Neon frame + corner brackets	Box-shadow border with matching glow · SVG corner bracket accents · frames the board like a real sign
■ Section neon dividers	Gradient horizontal rules with glow shadow · each section's divider matches its assigned colour

Theme Builder — Full (Pro tier)

■ 5 built-in preset themes	Bar Classic · Violet Lounge · Hot Summer · Ocean Dive · Rose Gold · sets all values at once
■ Title colour + glow pickers	Separate pickers · glow should be a darker shade of title colour for depth
■ Section colours	One neon ink colour per menu column · independent pickers · up to 4 section colours

■ Glow intensity slider	0.2× subtle chalk marker → 2.0× full electric overload · scales all text-shadow layers proportionally
■ Venue name font — 6 options	Pacifico · Lobster · Righteous · Fredoka One · Bungee · Permanent Marker
■ Menu items font — 4 options	Caveat · Kalam · Patrick Hand · Permanent Marker · shown in actual item text in live preview
■ Board background	Colour picker + 6 quick swatches for common dark board tones
■ Noto font preloading	Noto Sans SC, KR, JP, Arabic preloaded from this phase · prevents CJK flash when Phase 12 ships

Split Layout (Pro tier)

■ Left half — hero photos	2–4 bestseller photos · large format · pulled from MenuItem.IsPopular items
■ Right half — text menu	Complete menu in text columns · all items · allergen badges · pricing
■ Adjustable split ratio	Admin slider: 40/60 or 50/50 · stored on Screen entity · per-screen setting
■ Best for	Mid-range casual restaurants wanting visual appeal and complete menu information

Daily Special Hero Layout (Pro tier)

■ One large featured item	Full-screen photo + name + description + price · maximum visual impact
■ Rotating secondary strip	3–4 smaller items below · auto-rotates every 8 seconds · configurable dwell time
■ 'Today Only' badge	Amber banner on the hero item · creates scarcity · drives impulse orders
■ Best for	Restaurants pushing a new special every day · replaces paper A-frame signs

08

Scheduling Engine

BUILD · Weeks 18–22 · **Milestone:** Menus run themselves — zero daily effort

Eliminates the most common daily pain point: manually switching the menu board. Set it once and it runs indefinitely. The `IHostedService` evaluator runs every 60s and pushes `ContentUpdated` via `SignalR` whenever a schedule change is due.

Meal Period Auto-Switch

■ Period definitions	Breakfast · Lunch · Afternoon · Dinner · Late Night · each with <code>TimeOnly</code> start and end · stored per venue
■ Day-of-week control	Each period active on configurable days · e.g. Breakfast Mon–Fri, Brunch Sat–Sun
■ Layout per period	Each period can switch to a different layout, apply a menu filter, or change the active theme
■ Venue timezone evaluation	<code>ScheduleEvaluator</code> converts UTC to venue's IANA timezone before comparing · stored on Venue entity
■ Zero staff action	Once configured, no operator needs to touch the board between periods — ever

Happy Hour Scheduler (Pro tier)

■ Time window config	Start time + end time + active days · e.g. 4pm–7pm Mon–Fri · stored on <code>HappyHourSchedule</code> entity
■ Per-item HH pricing	<code>MenuItem.HHPrice</code> shown automatically during active window via <code>isHappyHour=true</code> in content payload
■ Auto-activation	<code>ScheduleEvaluator</code> pushes <code>isHappyHour=true</code> at start time · false at end · no manual trigger
■ Manual override	Admin can force-activate or force-deactivate regardless of schedule · useful for events
■ HH banner on display	Pulsing 'HAPPY HOUR · 4PM–7PM · MON–FRI' banner shown on board during active window
■ Countdown timer widget	'Happy Hour ends in 47 min' · live countdown · proven to drive last-minute orders

Playlist Rotation

■ Multiple slides per screen

Screen rotates through a defined slide list · configurable dwell time per slide in seconds

■ Slide types

Menu board · Daily special · Event promo · Image-only · Custom HTML (Phase 15)

■ Schedule per slide

Each slide has its own optional time window · only appears during its configured period

■ Drag-to-reorder

Admin reorders slides by dragging · live preview shows rotation order instantly

Emergency Broadcast

■ One button, all screens

'Cash only tonight' · 'Kitchen closing in 30 mins' · SignalR push reaches all screens instantly

■ Full-screen override

Broadcast replaces the current layout entirely · not a banner overlay

■ Auto-expire

Set a duration in minutes · board returns to normal layout automatically

■ Scope control

Push to all venue screens or target a specific screen

Date-Range Promotions

■ Start + end date

Seasonal specials with defined calendar dates · auto-activates and auto-expires

■ Christmas menu example

Activates Dec 1 · reverts Dec 27 · configured once, runs automatically

■ Event-specific layouts

Valentine's Day menu · Super Bowl specials · St Patrick's Day board · any date-bounded display change

09

Tap List Boards — Breweries & Bars

BUILD · Weeks 20–24 · **Milestone:** Can sell to breweries and taprooms

Three distinct tap list styles covering the full range of brewery and bar aesthetics. TapItem is a separate entity from MenuItem — beer-specific fields make the data model honest. Each layout uses the TapItem fields differently, as shown in the table below.

TapItem Field Usage Across Layouts

Field	Classic Chalk Board	Tap Strips Board	Digital Tap Board
Name	Column list item	Large hand-lettered	Bold card header
Style	Not shown	Small sub-label	Subtitle e.g. 'West Coast IPA'
ABV	Not shown	Small mono text	Shown with % suffix
IBU	Not shown	Not shown	Shown beside ABV
Description	Not shown	Not shown	2-line clamp tasting notes
Price	Category price only	Below style label	Top-right of card
GlassColor	Not used	Not used	SVG pint glass liquid fill colour
NameColor	Not used	Neon glow colour	Not used
Available	Greys list item	Hides or dims strip	Greys card to 40%

Classic Chalkboard Drinks Board

■ Purpose	Cocktail bars and bars with a fixed beer list · category pricing model rather than per-item prices
■ Drinks title	Large Pacifico header with blue neon glow · authentic chalk aesthetic
■ Category pricing	All cocktails \$10.95 · Import Beer \$4.00 · Domestic \$3.00 · prices in bordered boxes per category
■ Two-column cocktail list	Items in warm gold Caveat font · two columns · no individual prices shown

■ Beer sub-sections

Import and Domestic beers as named lists with bullet-dot separators

■ Chalk art illustrations

SVG cocktail glasses and beer bottle in chalk style · centred between sections

Tap Strips Board

■ Purpose

Taprooms and craft breweries · each keg gets its own strip · authentically brewery-style

■ 3-column grid

Each strip is a dark panel · tap number top-right · strips arranged in grid

■ Hand-lettered name

Each tap uses a rotating font from [Permanent Marker, Bungee, Righteous, Pacifico, Caveat] by index

■ Name glow colour

Each tap glows in its NameColor · neon text-shadow stack from Phase 07

■ Style + ABV

Caveat font · smaller and muted · supports the name without competing

■ Draw-in animation

Strips animate in sequentially left-to-right, top-to-bottom on load

Digital Tap Board

■ Purpose

Modern taprooms wanting a polished digital look · direct Taplist.io competitor

■ Wood texture

SVG fractal noise over warm dark brown + repeating-linear-gradient grain lines at 18% opacity

■ Beer glass SVG

Hand-drawn pint glass · liquid colour = GlassColor · foam drawn as white ellipses

■ Two-column card grid

2 columns × 3 rows = 6 taps · each card shows name, style, ABV, IBU, description, price

■ 'Now Brewing' callout

Coming-soon taps shown with amber badge · builds anticipation

Pairing Code Registration — Phase 09 Addition

■ Why here

Brewery customers have multiple TVs · typing URLs on TV remotes is the most-cited setup frustration

■ TV loads <code>vennu.app/pair</code>	Fullscreen display of 6-digit code + plain-language instructions · no URL to type
■ TV polls every 3 seconds	GET <code>/api/screens/pairing/{code}/status</code> · returns <code>{linked: true, screenId}</code> when admin claims
■ Admin enters code	POST <code>/api/screens/pairing/{code}/claim</code> · links screen · TV auto-redirects to its display URL
■ Code expiry	10-minute expiry · regenerates automatically · prevents stale codes from wrong venue claims

10

Upgrade Prompts & Billing UX

BUILD · Weeks 20–28 · **Milestone:** Self-serve upgrade funnel generating revenue

The in-product experience that turns trials into paying customers and paying customers into higher tiers. Every prompt is non-invasive — customers see what they're missing without their current workflow being interrupted or blocked.

Core Principles

Principle	Rule
Show benefit, not tier name	'Items sell out → board updates in seconds' — NOT 'Upgrade to Pro for POS integration'
Never block a workflow	Locked features are visible and informative · never an error or dead end
One prompt per screen	Never show more than one upgrade suggestion simultaneously · prevents alert fatigue
All prompts dismissible	Per-session dismiss memory · dismissed hints never reappear in the same session
Upgrade in one click	Prompt → upgrade modal → single CTA → Stripe Checkout · maximum two taps total

Tier Badge

■ Small coloured pill	e.g. 'PRO' in amber beside any locked feature · informational, never alarming
■ Used on	Nav items · section headers · individual form fields · layout cards
■ Colour matches tier	Starter=slate · Restaurant Starter=green · Pro=amber · Business=purple

Locked Nav Items

■ Visible at 50% opacity	Feature exists in sidebar · customers know it's there · not hidden from view
■ Tier badge beside label	Clear at a glance which tier unlocks this feature
■ Clicking opens modal	Not an error · upgrade modal opens with this specific feature as context

Locked Section Preview

■ Blurred glimpse	0.3px blur on a mockup screenshot of the feature · tantalising, not frustrating
■ One benefit sentence	Concrete and specific: 'Prices switch automatically at 4pm — no staff needed'
■ Soft CTA	Muted button · 'Unlock with Pro · \$89/mo' · not aggressive · easily ignored
■ No blocking behaviour	Customer continues using all unlocked features normally while preview is visible

Inline Feature Hint

■ Contextual placement	Appears at bottom of the most relevant panel · e.g. POS hint shown on menu editor tab
■ Blue left accent bar	Styled like an informational tip, not an advertisement
■ One per panel	Only the single most relevant locked feature shown per panel · never a list
■ Dismiss button	× closes for the session · hint never nags

Sidebar Nudge

■ Bottom of sidebar	Quiet and ambient · never in the way of main navigation
■ Rotates locked features	7-second intervals · dots indicate how many features are queued
■ Per-feature dismiss	Dismissing one rotates to the next · dismissed features never shown again
■ One sentence + one link	'Your team can update the board from anywhere · See how →'

Upgrade Modal — Bottom Sheet

■ Slides up from bottom	Native-feeling · not a pop-up · clicking backdrop dismisses
■ Feature name + benefit	What this specific feature does · concrete, tied to the feature clicked

■ All tier features as pills	Every feature at that tier shown as a pill · makes total value tangible
■ Current tier shown	'You're on Restaurant Starter' · context for the upgrade decision
■ Single CTA	'Upgrade to Pro — \$89/mo' · one button · goes directly to Stripe Checkout
■ 'Maybe later' text link	Easy non-guilt exit · customer remembers the feature when they're ready

Stripe Self-Serve Billing

■ Stripe Billing Portal	Pre-built hosted portal for plan changes and payment management · no custom UI needed
■ Upgrade flow	Upgrade modal CTA → Stripe Checkout with pre-selected plan → webhook fires → features unlock immediately
■ Trial conversion	Trial ends → Stripe invoice → grace period → if unpaid → past_due → features restricted
■ HaaS contract billing	Separate Stripe subscription per bundle · early cancel triggers buyout charge based on remaining term

11

POS Integration

PLAN · Months 5–8 (Weeks 24–36) · **Milestone:** Beats every generic signage competitor

Real-time POS sync is the single most powerful differentiator. When an item sells out at the register the board updates in under 500ms. Square is built first — best developer API, free sandbox, 4M+ restaurant customers, .NET SDK available.

Square — Build First

■ Why first	Best developer API · free sandbox · 4M+ restaurant customers · .NET SDK · Square Marketplace listing
■ OAuth 2.0 connect flow	'Connect Square' button in admin · one-click auth · access token encrypted in Azure SQL
■ Catalog API sync on connect	Full menu auto-imported from Square on first connect · zero manual data entry for the customer
■ Inventory webhook receiver	POST /webhooks/square · return HTTP 200 immediately · process in background Task.Run() · never times out
■ Price sync	Price change in Square POS → webhook → ContentUpdated or ItemAvailabilityChanged via SignalR → board updates <200ms
■ Async processing	Return HTTP 200 immediately · process in background Task · never times out on Square's side
■ Square App Marketplace	Free listing · gives access to 4M+ restaurant customers searching for display tools

Toast — Build Second

■ Why second	Dominates full-service restaurant market — the primary Pro tier customer profile
■ Toast developer sandbox	dev.toasttab.com · free sandbox for development and testing
■ Webhook registration	Not self-serve · submit production endpoint URL to Toast developer contact · requires approval
■ Hourly polling fallback	GET menu availability every 60 minutes as resilience · Toast recommends alongside webhooks

■ GUID deduplication

Log Stripe event IDs in ProcessedWebhookEvents table · idempotent handlers safe for duplicate events

Clover — Build Third

■ Why third

Fills mid-market gap between Square (SMB) and Toast (enterprise)

■ Clover REST API + OAuth

GET /v3/merchants/{merchantId}/items · same OAuth flow pattern as Square

■ Inventory update webhook

Subscribe to inventory change events via Clover Developer Dashboard

■ Same IPosProvider pattern

Third provider adds one class · no changes to Square or Toast code

Integration Architecture

■ Unified webhook endpoint

POST /webhooks/{provider} · one controller receives all three providers · routes to correct IPosProvider

■ IPosProvider interface

Clean .NET abstraction · fourth POS adds one class and nothing else changes

■ ItemAvailabilityChanged push

SignalR pushes to all venue screens within 500ms of POS event · player patches single item · no full re-render

■ Apideck evaluation

\$300–500/mo unified wrapper · evaluate when 50+ venues request less common POS systems simultaneously

12

Multilingual Support

PLAN · Months 7–9 (Weeks 32–40) · Milestone: Ethnic restaurant market unlocked

No signage competitor does multilingual well. MenuItemTranslation was created in Phase 02 and Noto fonts were preloaded in Phase 07 — this phase builds the editing UI and bilingual rendering on top of that foundation. No new migrations or font loading needed.

Admin UI Translation

■ react-i18next	Auto-detects browser language · language preference stored on User entity
■ Language switcher in header	Flag emoji + language name · instant switch without page reload · persists across sessions
■ Launch languages — 3	Spanish · Simplified Chinese · Vietnamese · highest-ROI non-English restaurant owner markets in the US
■ Phase 2 languages	Korean · Portuguese · add after validating demand from the initial three
■ All UI strings externalised	Zero hardcoded English text from Phase 05 · this phase only provides the translation JSON files

AI Bulk Translation — Claude API

■ One-click translate menu	'Translate to Chinese' button · Claude API call · all items translated in approximately 15 seconds
■ Cost	~\$0.30 to translate a 50-item menu into 3 languages · essentially zero cost at any customer volume
■ Context-aware	Claude understands restaurant terminology · 'Kung Pao' → '■' not a literal word-for-word translation
■ Review table	All translations shown side-by-side · owner edits any inline · marks as reviewed
■ IsAutoTranslated flag	DB flag distinguishes AI-generated from owner-entered text · used for quality tracking

Bilingual Display Modes

■ Stack mode	Primary language full size · secondary smaller below each item · cleanest option for most boards
■ Side-by-side mode	Two full columns one per language · equal visual weight · good for equal-prominence bilingual venues
■ Subtitle mode	Primary large · translation as small italic subtitle · minimal extra vertical space
■ Per-venue setting	Admin chooses mode · stored on Venue entity · applied across all layouts
■ Font pairing	English in Caveat/Kalam · CJK/Arabic in matching-weight Noto (preloaded since Phase 07)

RTL Support

■ Arabic and Hebrew	dir=rtl applied to RTL language content · layout mirrors via CSS logical properties
■ Noto Sans Arabic	Already preloaded from Phase 07 · no additional font loading time in this phase
■ Admin RTL mode	Admin UI can also render in RTL for Arabic and Hebrew-speaking venue owners

Screen Registration — Phase 12 Update

■ Language on pairing screen	venu.app/pair shown in browser's detected language automatically
■ Venue language inherited	New screens automatically inherit venue's primary and secondary language settings

13

Staff Mobile App

PLAN · Months 9–11 (Weeks 40–48) · **Milestone:** Pro tier becomes sticky

The single most-requested feature post-launch. A bar manager who can't find the laptop at 6pm Friday will cancel. The one who can update from their iPhone will refer Vennu to every venue they know. Included in Pro at no extra charge — it is a retention feature, not an add-on.

React Native — iOS + Android

■ One codebase	React Native · submitted to App Store and Google Play · no separate teams needed
■ Shared .NET API	Same endpoints as web admin · no new backend required · existing auth tokens work on mobile
■ Biometric authentication	Face ID / Touch ID · fast login for operators who check it mid-service
■ Push notifications	Firebase Cloud Messaging · screen offline alerts · keg blow · daily special reminders

Core Actions

■ Quick 86 toggle	Mark item unavailable in 2 taps from the kitchen · SignalR push · board updates in under 200ms
■ Daily special push	Text entry + push button · live on all screens in under 10 seconds
■ Happy hour override	Activate or deactivate instantly regardless of schedule · useful for events or early close
■ Emergency broadcast	'Cash only tonight' from the phone · all screens update immediately via SignalR

Screen Health

■ All screens at a glance	Online/offline status · last-seen timestamp · manual push button per screen
■ Offline alert notification	Push notification the moment any screen goes offline during service hours

■ Remote content refresh

Force any screen to reload its full content payload with one tap

Brewery-Specific**■ Keg blow notification**

Staff marks keg empty → board removes/greys that tap + push notification to manager

■ Tap list quick-edit

Change ABV, price, or description from the phone without opening a laptop

■ 'Now Pouring' update

Mark new keg as live · board updates the tap strip immediately via SignalR

14

TV Apps & Platform Distribution

PLAN · Months 11–15 (Weeks 48–64) · **Milestone:** App store presence on all major TV platforms

Thin native wrappers around the existing React display SPA. Player code is identical across every platform — only packaging and build target change. The pairing code flow from Phase 09 handles all setup with no URL typing on any platform.

Android TV + Amazon Fire TV

■ Why first	Fire TV Stick is the most common hospitality hardware · one APK covers both platforms
■ Kotlin WebView wrapper	Thin native shell · <code>webView.loadUrl('vennu.app/display/{screenId}')</code> · zero player code changes
■ Pairing code on first launch	App shows 6-digit code automatically · admin claims · TV redirects to display URL in 30s
■ Boot receiver	Android <code>BOOT_COMPLETED</code> intent · Vennu starts automatically when TV powers on
■ Kiosk / pinned mode	Device owner mode · prevents exit without PIN · always-on display
■ Google Play for TV	Vennu searchable in TV app store · customers install like Netflix · no sideloading
■ Amazon Appstore submission	Same APK · separate submission · covers all Fire Stick variants including 4K Max

Samsung Tizen

■ Why important	Samsung dominates commercial display market · app store listing removes all setup friction
■ Tizen web app package	React SPA + <code>config.xml</code> manifest · submitted to Samsung Smart Signage Platform (SSSP)
■ Zero player changes	Same display app code · Tizen packages and hosts it · pairing code works identically
■ Samsung B2B program	Enables pre-installation on Samsung screens shipped as part of HaaS bundles

LG webOS

■ IPK package	Same React SPA · different build target · submitted to LG Smart Signage Solution (SSS)
■ webOS simulator	Free LG developer tool for Mac/Windows · test without physical hardware
■ Coverage	LG is the second-largest commercial display brand globally

Screen Registration — Phase 14 Update

■ TV app pairing screen	App shows code fullscreen on first launch · vennu.app/pair instruction text below
■ Auto-redirect on claim	TV transitions seamlessly from pairing screen to content display when claimed
■ Platform field	Screen.Platform set to 'android_tv' / 'fire_tv' / 'tizen' / 'webos' · visible in Super Admin
■ App version tracking	Player reports app version in each heartbeat ping · Super Admin flags outdated screens

HaaS Pre-Registration

■ Screens arrive pre-configured	Create screen records before shipping · venue content loaded before the box is sealed
■ Customer experience	Powers on the TV · correct menu already showing · zero setup required
■ Card in the box	'Your screens are already set up. Plug in, power on, done.'
■ No pairing code needed	Screens registered to venue before delivery · shows Online after first heartbeat

15

AI Features

PLAN · Months 14–18 (Weeks 60–76) · Milestone: Premium tier upsell driver

AI that lowers the skill floor for every venue owner. A restaurant operator who has never written a menu description gets professional copy in seconds. All features use the Claude API at approximately \$0.01–0.03 per call. AI bulk translation was introduced in Phase 12 — this phase adds all remaining AI capabilities.

AI Menu Content — Claude API

■ Menu description writer	'grilled salmon, lemon butter' → 'Pan-seared Atlantic salmon with house-made lemon beurre blanc'
■ Bulk description generation	Generate descriptions for every item missing one · one click · review table before publishing
■ Smart naming suggestions	Highlights generic names like 'Chicken Dish' · suggests specific alternatives · owner accepts or edits
■ Allergen detection	Claude reads description · suggests relevant allergen badges · owner confirms each before applying

AI Custom Display Builder (Business tier)

■ Plain English input	'Dark blue, gold text, logo top centre, beer list left, food right, countdown bottom right'
■ Claude generates layout	Full JSX/CSS generated · saved to Azure Blob as a named reusable template
■ Template variables	{{venueName}} · {{items}} · {{sections}} · {{isHappyHour}} · {{currentTime}} injected at render time
■ iframe sandbox	sandbox='allow-scripts' · security boundary prevents XSS · SignalR still pushes live content updates
■ Saved and reusable	Generated templates persist · owner regenerates with a new prompt anytime
■ ~\$0.02 per generation	Business tier only · usage tracked per venue · metered via TierFeature.LimitValue

HTML/CSS Sandbox Editor (Developer tier)

■ Monaco Editor

VS Code engine in the browser · syntax highlighting · autocomplete

■ Variable reference panel

Sidebar lists all {{variables}} with live examples using real venue menu data

■ Real-time preview

Renders with live menu data while editing · no save needed · immediate feedback

■ Not shown to standard owners

Developer tier feature for technical operators and white-label agency resellers

Smart Features

■ Happy hour window suggester

Analyses POS order volume by hour · recommends optimal slow-period windows · one-click apply

■ Auto-layout position advisor

Compares display position with POS sales · suggests moving high-margin items to prime positions

■ Photo background generator

Upload a plain food photo · AI generates a styled background matching venue brand colours

16

Analytics & Smart Features

PLAN · Months 17–22 (Weeks 72–92) · Milestone: Data-driven venues — clear ROI proof

Analytics needs real usage data and POS history to be meaningful. By Phase 16 there are months of heartbeat logs, impression data, and POS correlation. Empty charts at launch would have zero value and damage customer trust in the feature.

Analytics Dashboard

■ Screen uptime tracking	Heartbeat pings logged per screen · uptime % per week · downtime periods on timeline
■ Content impression logging	Which items displayed · when · for how long · total daily impressions per item
■ POS sales correlation	Display position vs. actual POS sales · powered by Square/Toast integration data
■ Item performance scoring	'Your #2 revenue item sits at screen 3, position 8 — consider moving it to screen 1, position 1'
■ Happy hour ROI report	Order volume and revenue before, during, and after happy hour · quantified financial impact
■ Multi-location rollup	Business tier only · aggregate reporting across all branches · compare venue performance

A/B Layout Testing

■ Test two layouts	Layout A Mon/Wed · Layout B Tue/Thu · metric is average order value from POS
■ Item position testing	Test item in position 1 vs position 4 · measure POS sales impact over 2 weeks
■ Automatic winner	System identifies winner after configurable period · prompts admin to make it permanent

Dynamic Pricing (Business tier)

■ Time-of-day pricing	Lunch specials at \$9.99 auto-activate at 11am · revert at 3pm · ScheduleEvaluator driven
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■ Rules engine

Admin sets conditions: 'If weekday AND hour < 17:00 THEN use lunch_price field' · JSON rules

■ Demand-based pricing

Slower periods get discounted prices automatically · yield management for sophisticated operators

Smart Context Features

■ Foot traffic mode

Google Maps Popular Times API · simplified board during peak rush hours

■ Weather-triggered promotions

OpenWeather API · temperature > 85°F → cold drink special activates automatically

■ Social proof ticker

Google and Yelp review rotation · fetched via Places API · shown at point of decision

■ Upsell prompt engine

'Pairs well with our Hazy IPA' shown beneath featured items · configured per-item in admin

■ Live sports ticker

Footer bar for sports bars · team scores via sports API · keeps customers engaged

Webhook & Automation

■ Outbound webhooks

Vennu fires events to customer's own systems when content changes · POST to their configured endpoint

■ Zapier integration

Connect to any web service · Google Sheets price sync · Slack screen-offline alerts

■ Google Sheets sync

Owner updates a price in Sheets · Zapier → Vennu webhook → board updates automatically

Revenue Model

Software-Only Tiers

Tier	Price	Screens	Key Features
Starter	\$39/mo	2	Photo Grid · Classic Diner · basic scheduling · allergen badges · bas
Restaurant Starter	\$49/mo	1	Above + meal periods · bilingual · AI translation (1 lang) · Quick Upd
Pro	\$89/mo	6	All layouts · happy hour · POS · staff app · video wall · full analytics
Business	\$179/mo	Unlimited	All Pro + AI custom builder · multi-location · white-label · HTML edito

Annual billing: 2 months free (~17%). 14-day free trial with no credit card on all tiers.

Hardware-as-a-Service Bundles

Bundle	Monthly	Term	Includes	Post-Contract
Starter Kit	\$89/mo	18-mo	1 screen + Fire Stick	Auto → \$39 Starter
Bar Pack	\$159/mo	24-mo	2 screens + mounting + sticks	Auto → \$89 Pro
Full House	\$249/mo	36-mo	4 screens + mounting + setup visit	Auto → \$179 Business

Hardware ownership transfers at contract end. Stripe auto-transitions to software tier. HaaS LTV over 5 years: Full House = ~\$13,000 vs ~\$1,600 software-only (8× more valuable).

MRR Milestones — Pro @ \$89/mo

Customers	MRR	Annual	Significance
4	\$356	\$4,272	Azure + Stripe fees break-even
10	\$890	\$10,680	Ramen profitable
65	\$5,785	\$69,420	Break-even with \$5,000/mo founder salary
100	\$8,900	\$106,800	Healthy single-founder business
200	\$17,800	\$213,600	First hire justified

500	\$44,500	\$534,000	Series-A territory
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